

FMUSER FUQ-4427 IP QAM Modulator 512 IP Inputs-16 QAM Outputs GE Ports DVB-C J.83 A-B Multiplexing Scrambling CATV Headend System Product Overview



The FMUSER FUQ-4427 is a 16-channel IP QAM modulator integrating multiplexing, scrambling, and modulating functions. It accepts up to 512 IP inputs via three GE ports and outputs 16 non-adjacent RF carriers compliant with DVB-C and ITU-T J.83 A/B standards. Managed through web-based network management, it enables easy configuration and monitoring. Ideal for CATV broadcasting and digital headend applications.

Key feature

Supports up to 512 IP inputs through three GE ports with 840Mbps per port.

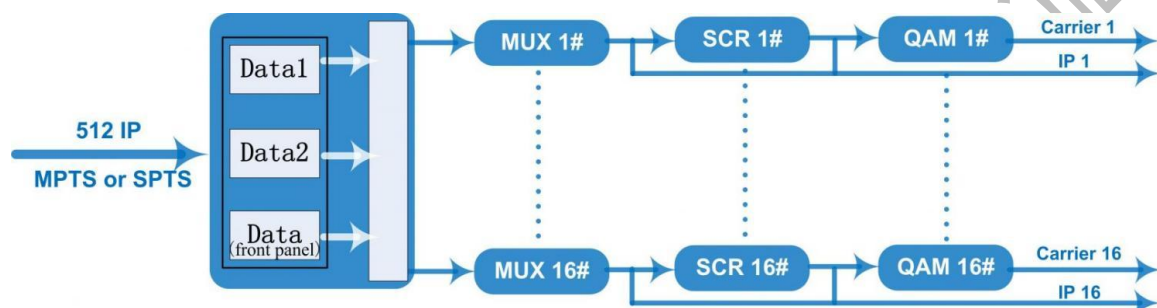
Delivers 16 non-adjacent RF carriers compliant with DVB-C (EN 300 429) and ITU-T J.83 A/B standards.

Features DVB scrambling systems including ETR289, ETSI 101 197, and ETSI 103 197.

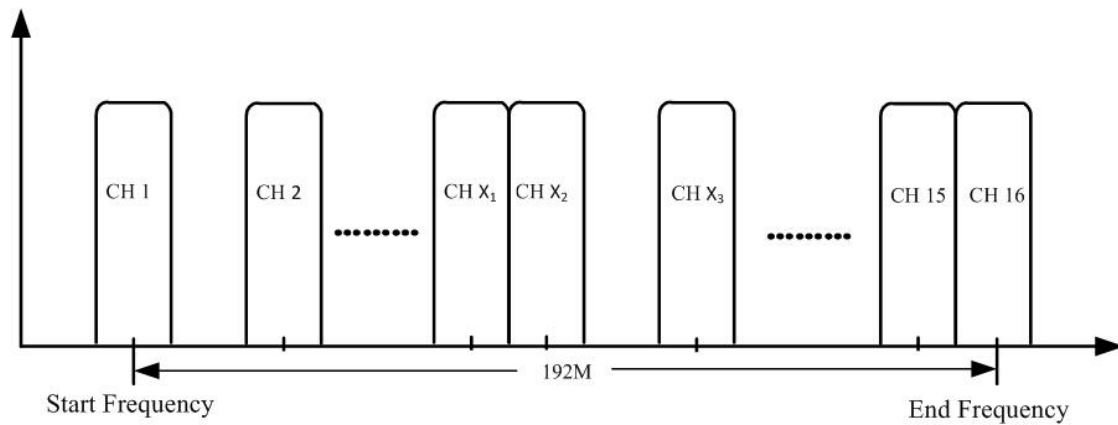
Enables web-based network management for configuration, monitoring, and control.

Supports PID remapping (up to 180 per channel), CA filtering, and PSI/SI editing.

Inner Principle Chart



Carrier Setting Illustration



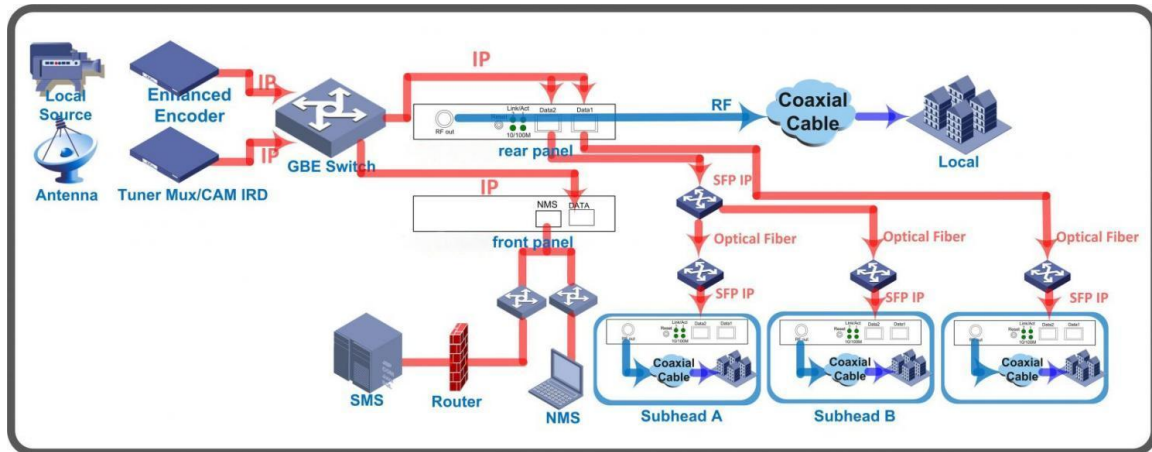
Specifications

Input	Input	Max 512 IP input through 3 (front-panel Data port, Data 1 and Data 2) 100/1000M Ethernet Port (SFP interface optional). Each Data1 or Data 2 port can input max 512 IP, while front-panel Data port can input max 128 IP	Max 512 IP input through 3 (front-panel Data port, Data 1 and Data 2) 100/1000M Ethernet Port (SFP interface optional). Each Data1 or Data 2 port can input max 512 IP, while front-panel Data port can input max 128 IP
Input	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Transmission Rate	Max 840Mbps for each GE input	Max 840Mbps for each GE input
Mux	Input Channel	512	512
Mux	Output Channel	16	16
Mux	Max PIDs	180 per channel	180 per channel
Mux	Functions	PID remapping(auto/m annually optional)	PID remapping(auto/m annually optional)
Mux	Functions	PCR accurate adjusting	PCR accurate adjusting
Mux	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling Parameters	Max simulcrypt CA	4	4

Scrambling Parameters	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Connection	Local/remote connection	Local/remote connection
Modulation Parameters	DVB-C Modulator Section	J.83A	Constellation : 16/32/64/128/256QAM
Modulation Parameters	DVB-C Modulator Section	J.83A	Bandwidth :8M
Modulation Parameters	DVB-C Modulator Section	J.83B	Constellation : 64QAM/ 256QAM
Modulation Parameters	DVB-C Modulator Section	J.83B	Bandwidth :6M
Modulation Parameters	QAM Channel	16 non-adjacent carrier	16 non-adjacent carrier
Modulation Parameters	Modulation Standard	EN300 429/ITU-T J.83A/B(DVB-C)	EN300 429/ITU-T J.83A/B(DVB-C)
Modulation Parameters	Symbol Rate	5.0~7.0Msps, 1ksps stepping	5.0~7.0Msps, 1ksps stepping
Modulation Parameters	FEC	RS (204, 188)	RS (204, 188)
RF Output	Interface	1 F typed output port for 16 carriers, 75Ω impedance	1 F typed output port for 16 carriers, 75Ω impedance
RF Output	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output	Output Level	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping
RF Output	MER	≥ 40dB	≥ 40dB
TS output	16 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports	16 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports	16 IP output over UDP/RTP/RTSP, unicast/multicast, 2 (Data1& Data2) 100/1000M Ethernet Ports
System	web-based Network management	web-based Network management	web-based Network management
General	Demission	420mm×440mm×44.5 mm (WxLxH)	420mm×440mm×44.5 mm (WxLxH)

General	Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)
General	Power Supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz

System Connection Sample (IP Based)



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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